

The last two weeks I have worked a lot with materials to make sensors in the FYRE program. The first day of class we were tasked with making sensors that could detect water vapor, and were given only a sheet of paper and an ammeter. To solve this problem, my group and I first considered semiconductive materials, as resistance changes significantly with contact from another material. We chose graphite as the material as we were able to extract some of it from our pens. We noticed when the water covered the graphite, the combination of these things lowered the resistance.

The next few days of class I got the chance to test out resistivities of different materials, mainly silver and copper. I learned the connection between resistance and resistivity, and how to run my own tests to find resistivity of any material. In order to find resistivity, my group and I would take a long strand of a material with a small cross sectional thickness. Using dial calipers, we would measure the cross sectional area through finding the height and the width of the material and measure the length. Once we got these measurements down, we would use the ammeter to measure resistance, performing some calculations of resistance times area divided by the length would give us the answer for resistivity.

My group and I then patterned our own electrodes using silver and glass to screen print. It involved using our vinyl cut as a pattern for the paste, and then using the hot plate to dry it. Finally, I learned sensor applications such as photolithography and spincoating. For photolithography, I visited the clean room and learned about tiny electrodes that are patterned on silicon wafers. With spincoating, I learned about a material called PEDOT-PSS, a conductive polymer used in sensors.